

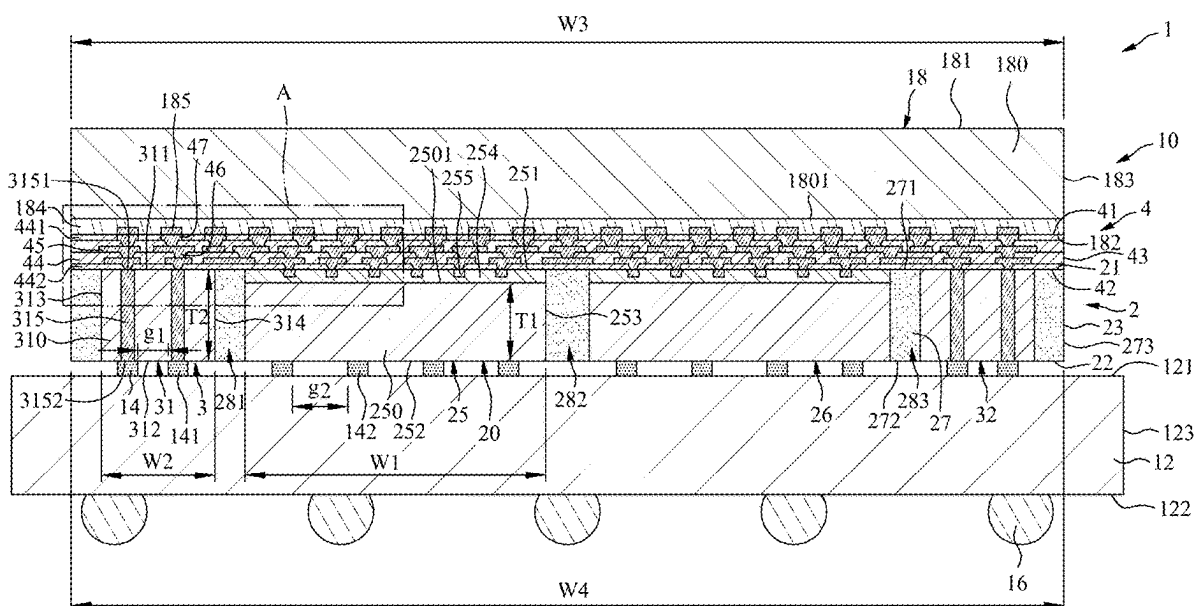
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H01L 23/28 (2006.01)

A package structure, an assembly structure and a manufacturing method are provided. The package structure includes at least one semiconductor device, at least one inactive element and an encapsulant. The at least one inactive element is disposed around the at least one semiconductor device, and includes a main portion and at least one through via extending through the main portion. The encapsulant encapsulates the at least one semiconductor device and the at least one inactive element. A first surface of the encapsulant is substantially coplanar with a first surface of the at least one inactive element and a first surface of the at least one semiconductor device. A second surface of the encapsulant is substantially coplanar with a second surface of the at least one inactive element and a second surface of the at least one semiconductor device.



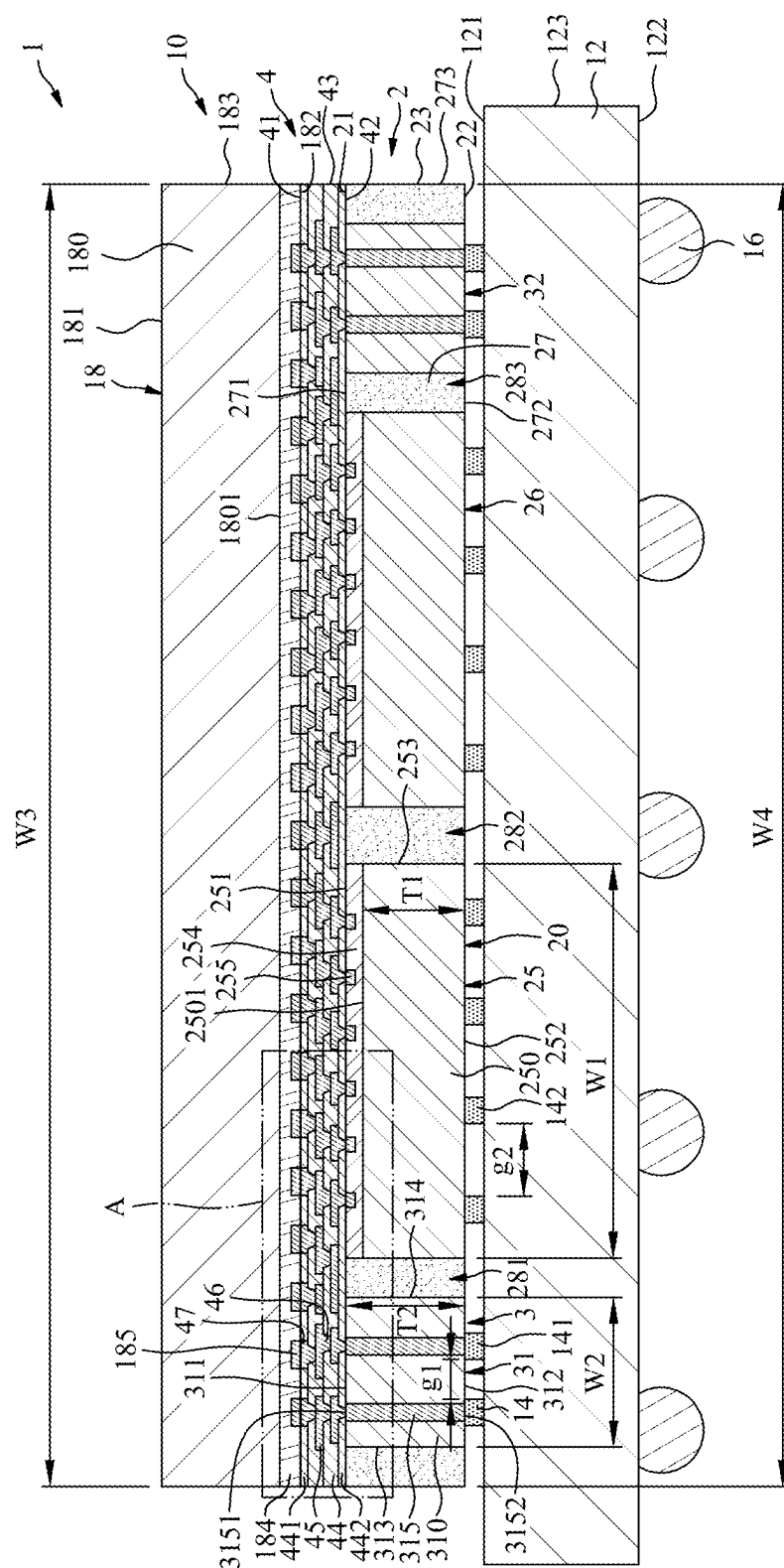


FIG. 1

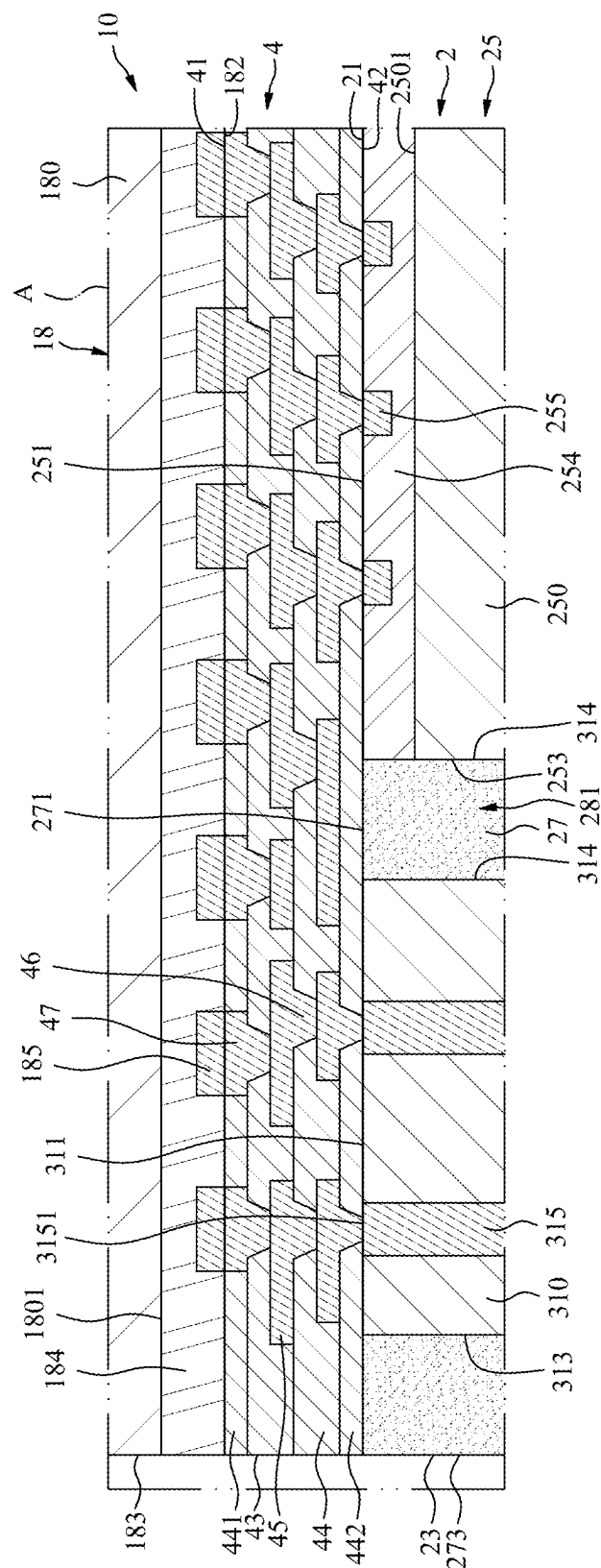


FIG. 2

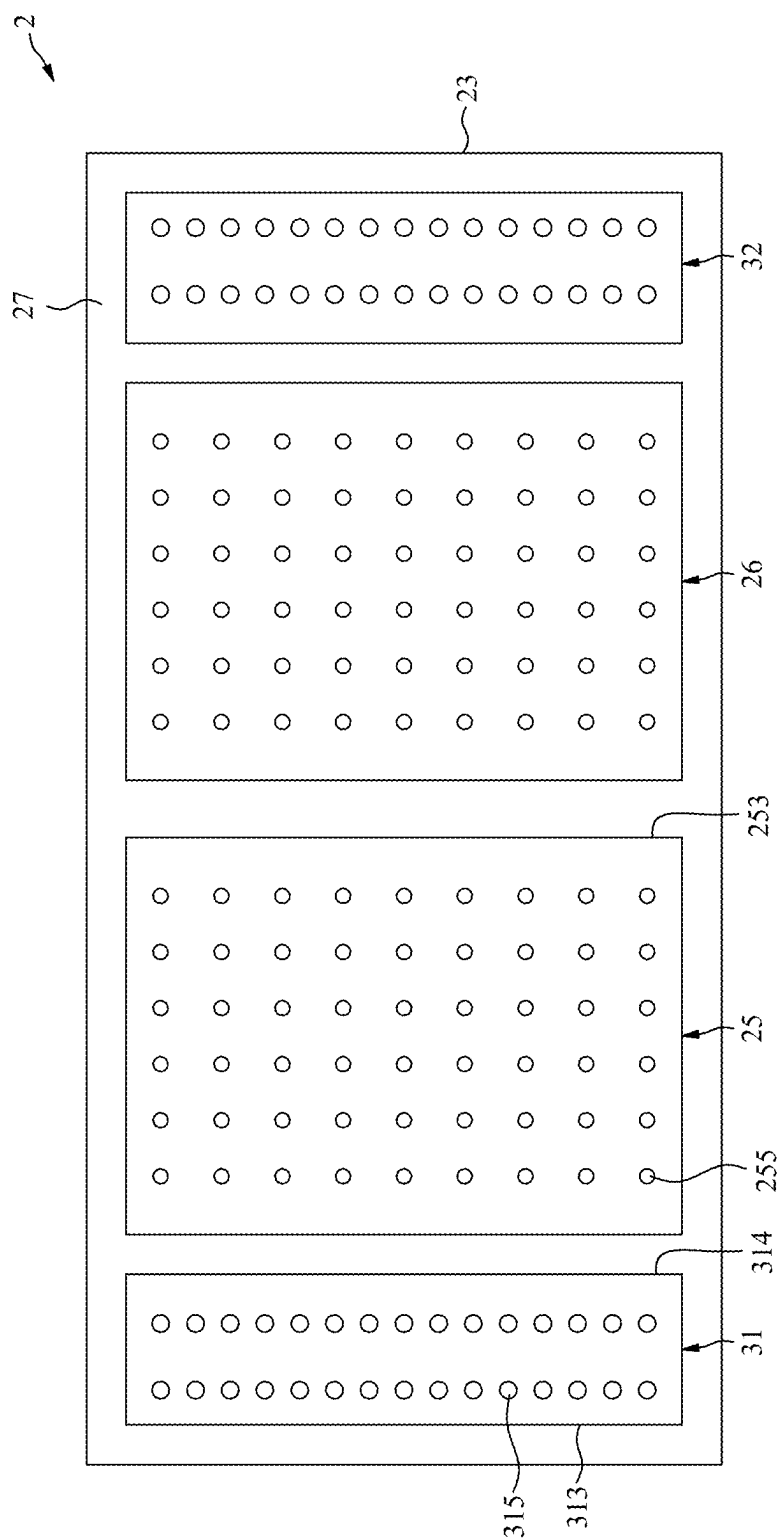


FIG. 3

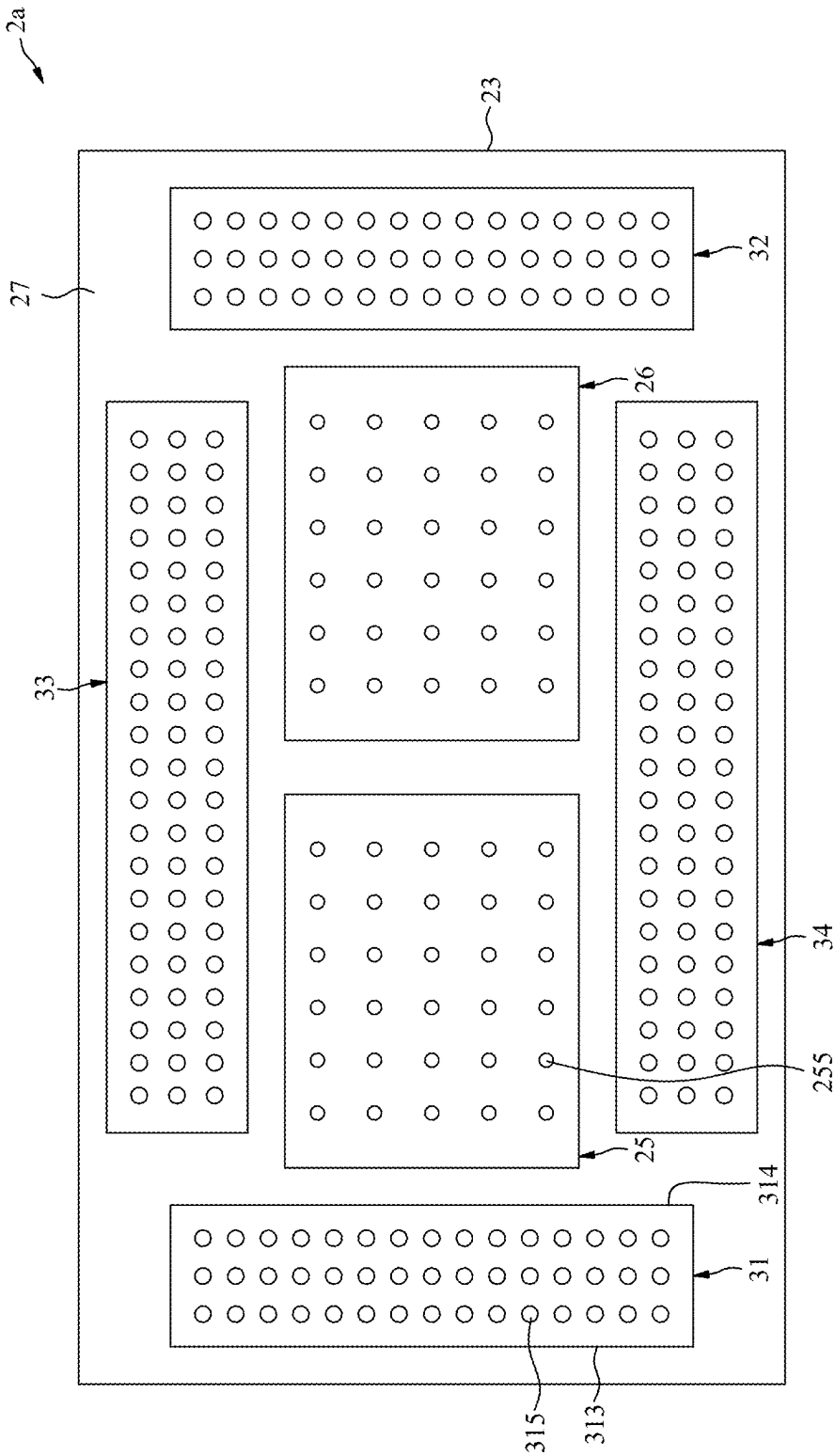


FIG. 4

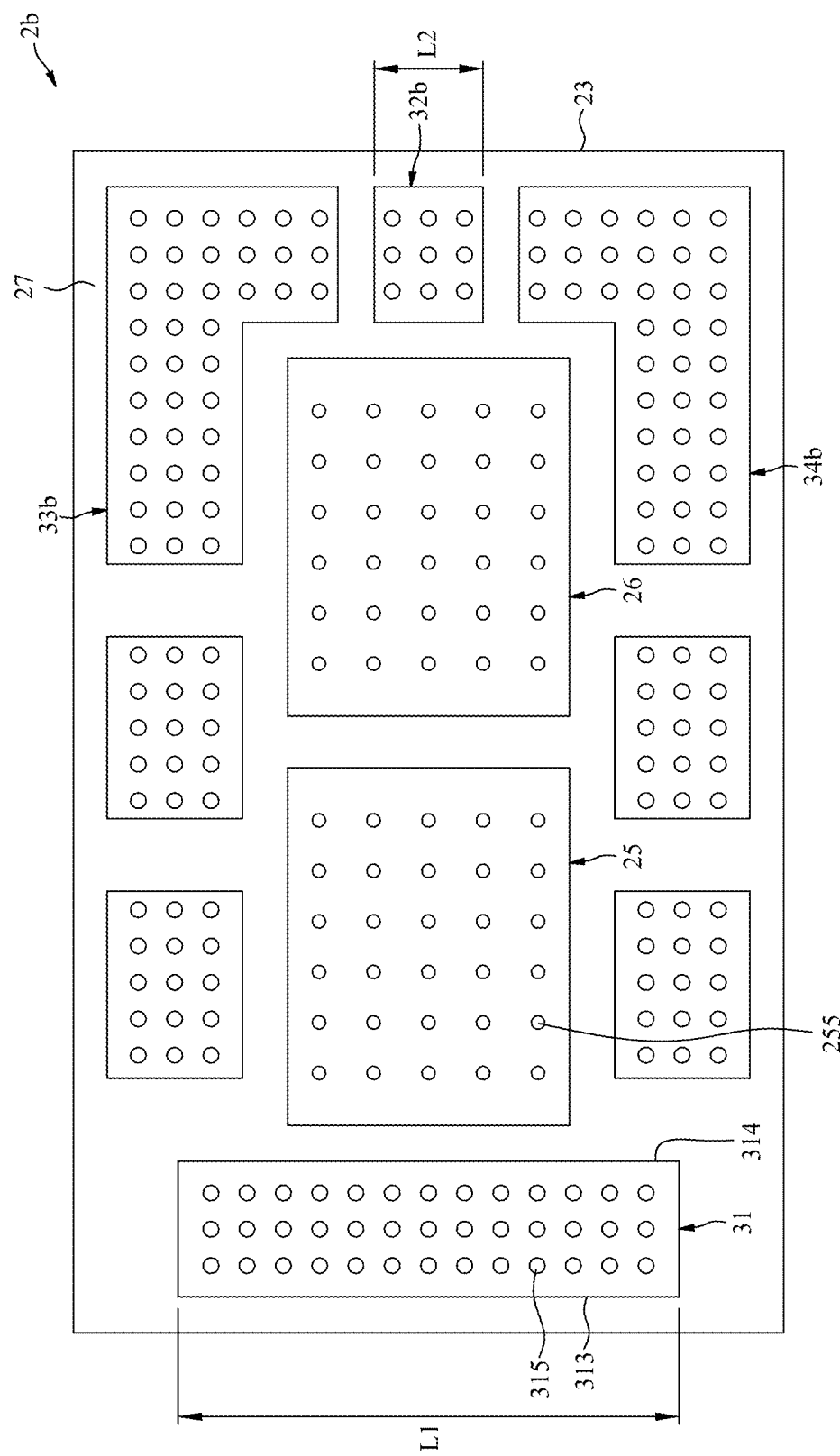


FIG. 5

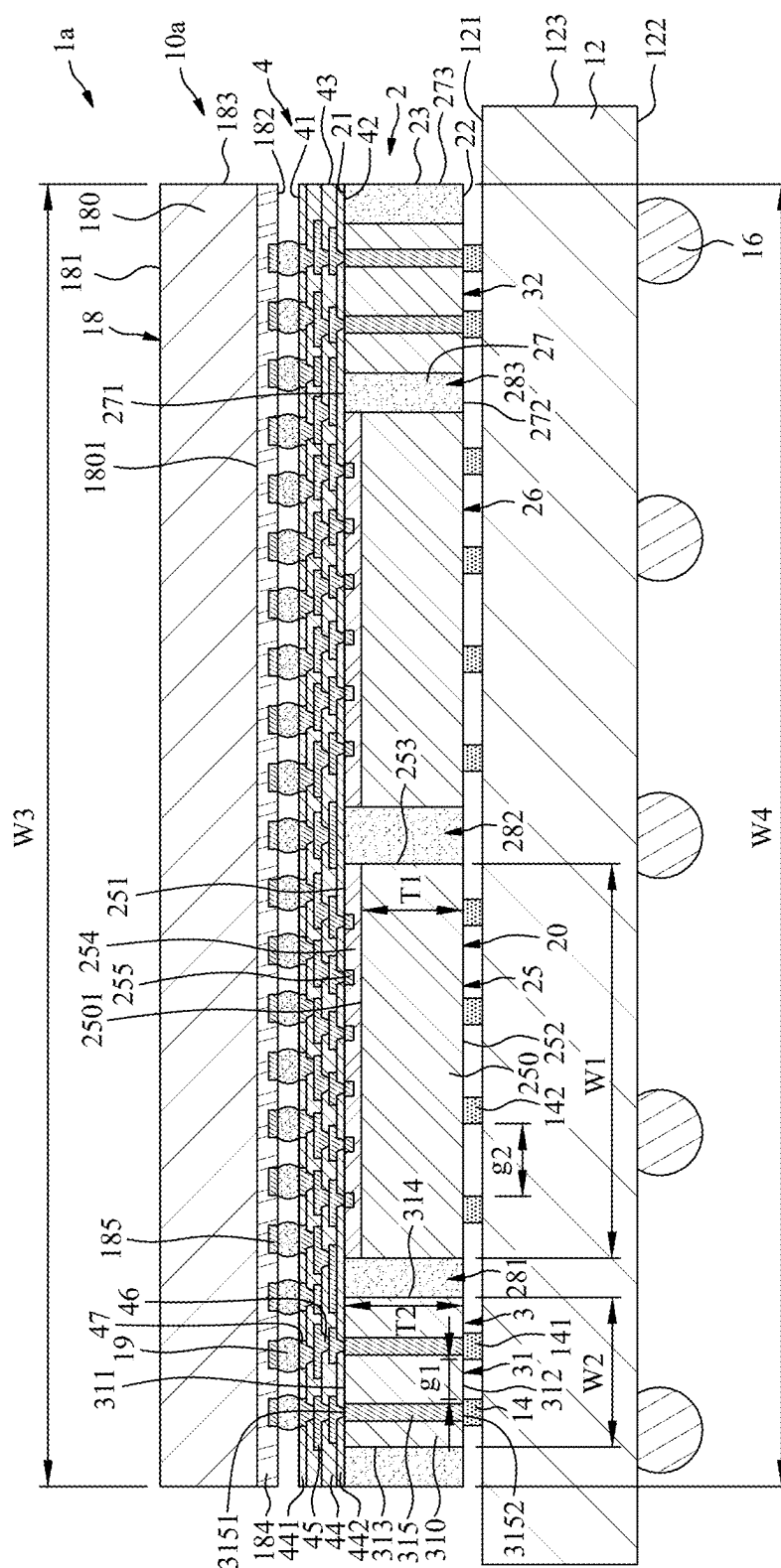


FIG. 6

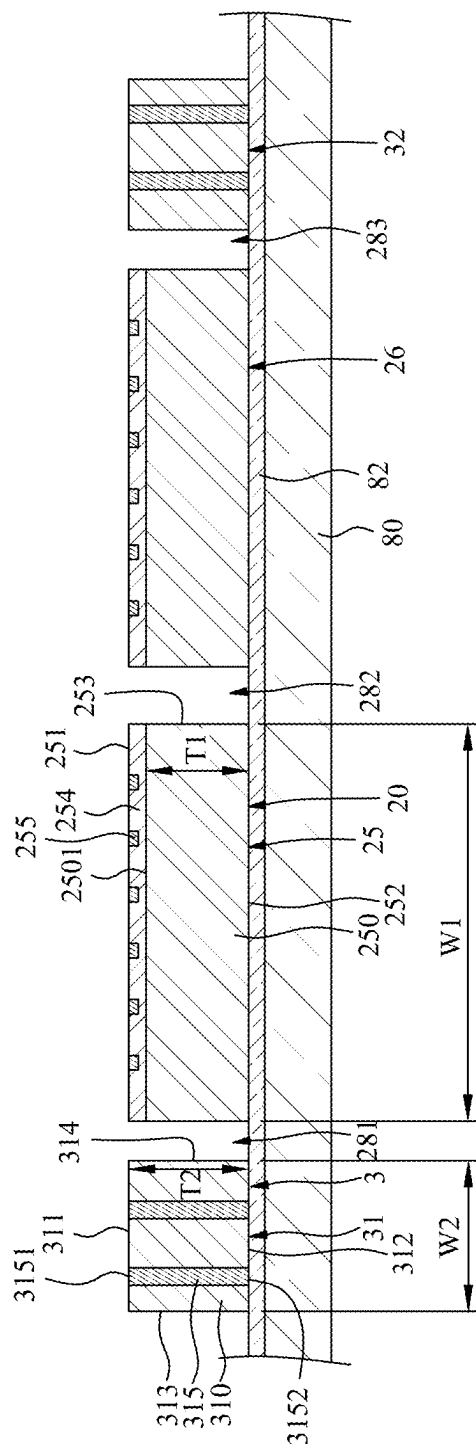


FIG. 7

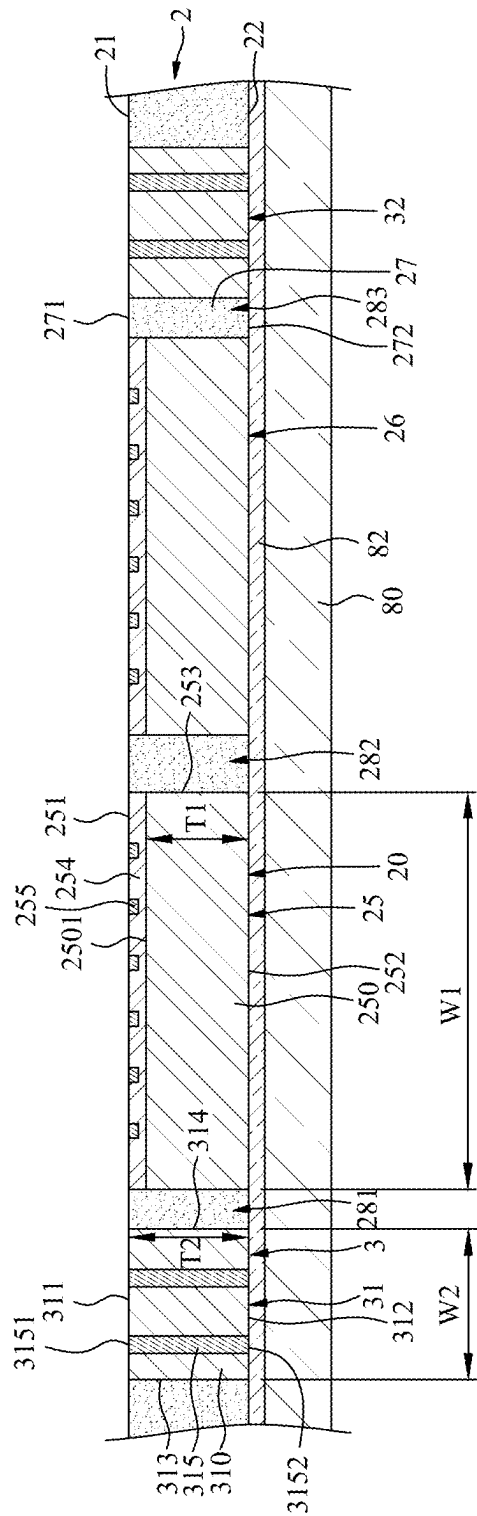


FIG. 8

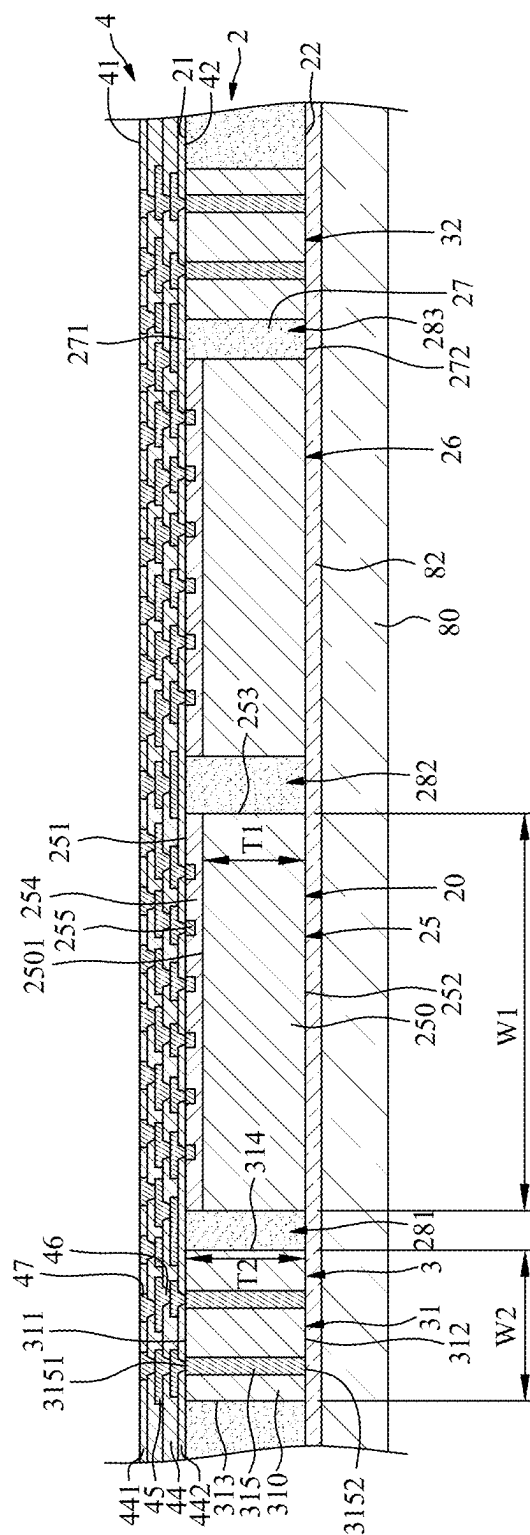


FIG. 9

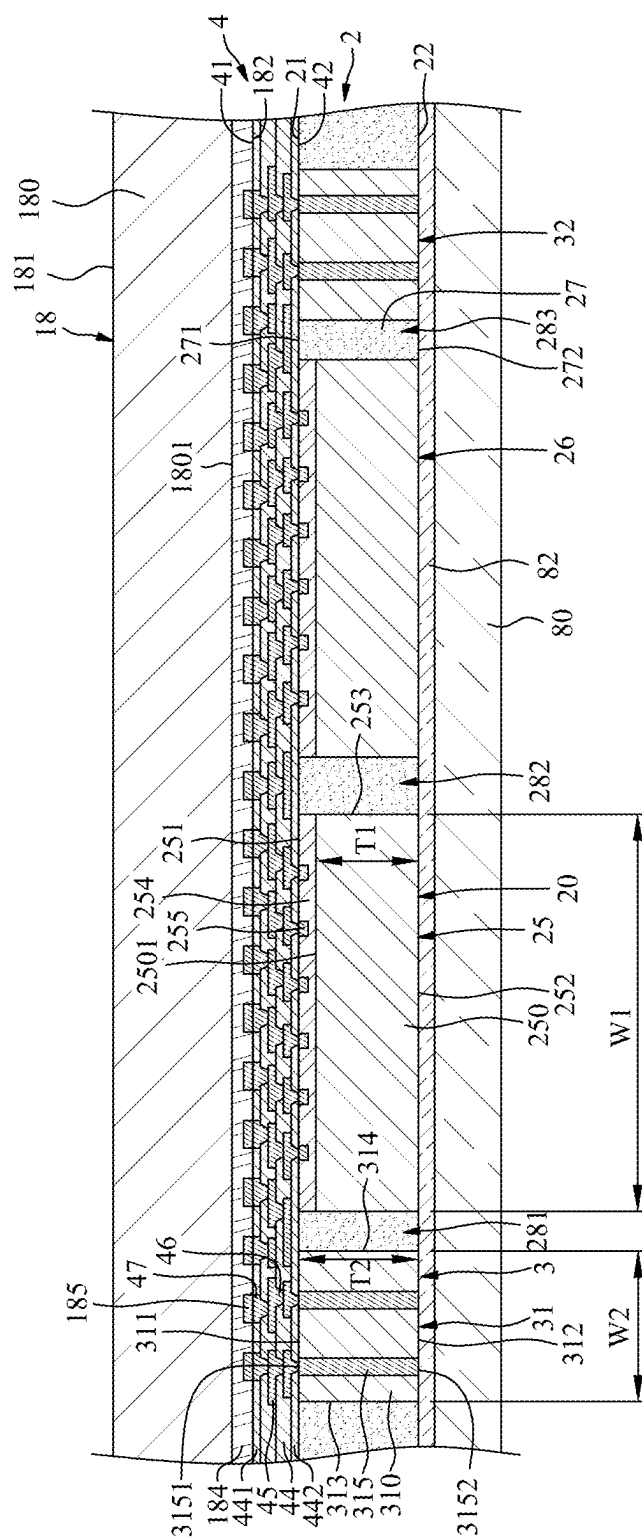


FIG. 10

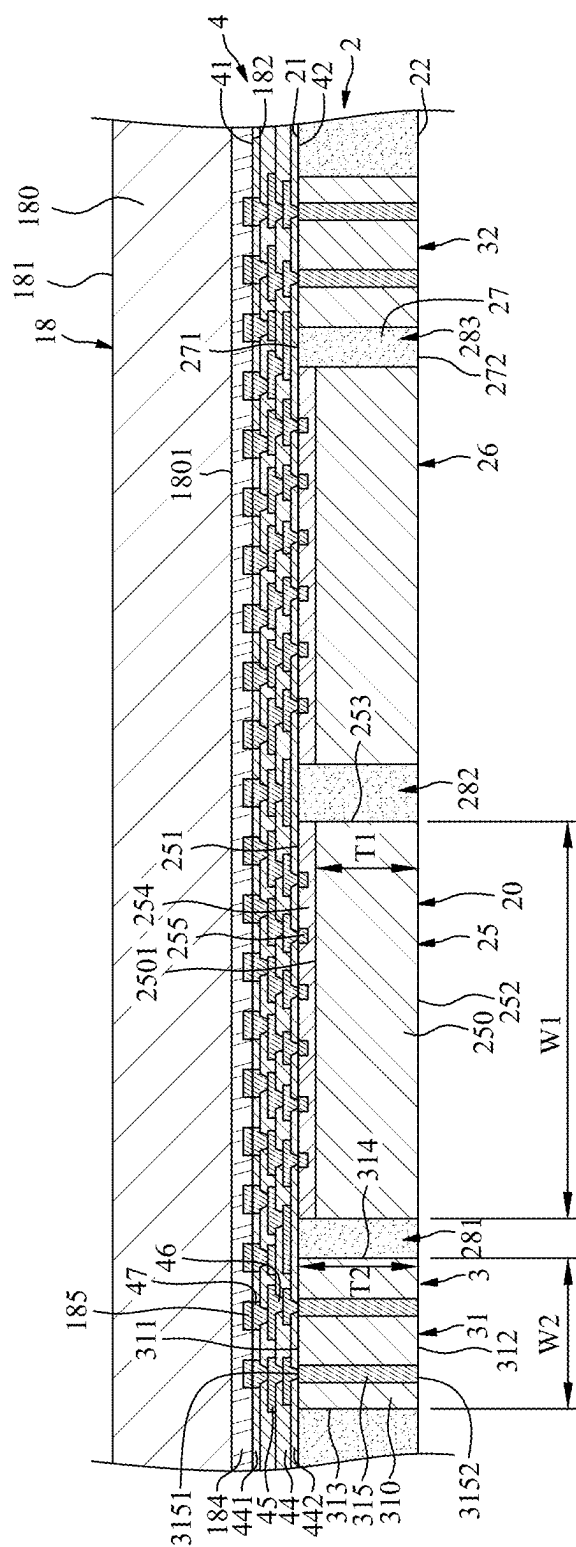


FIG. 11

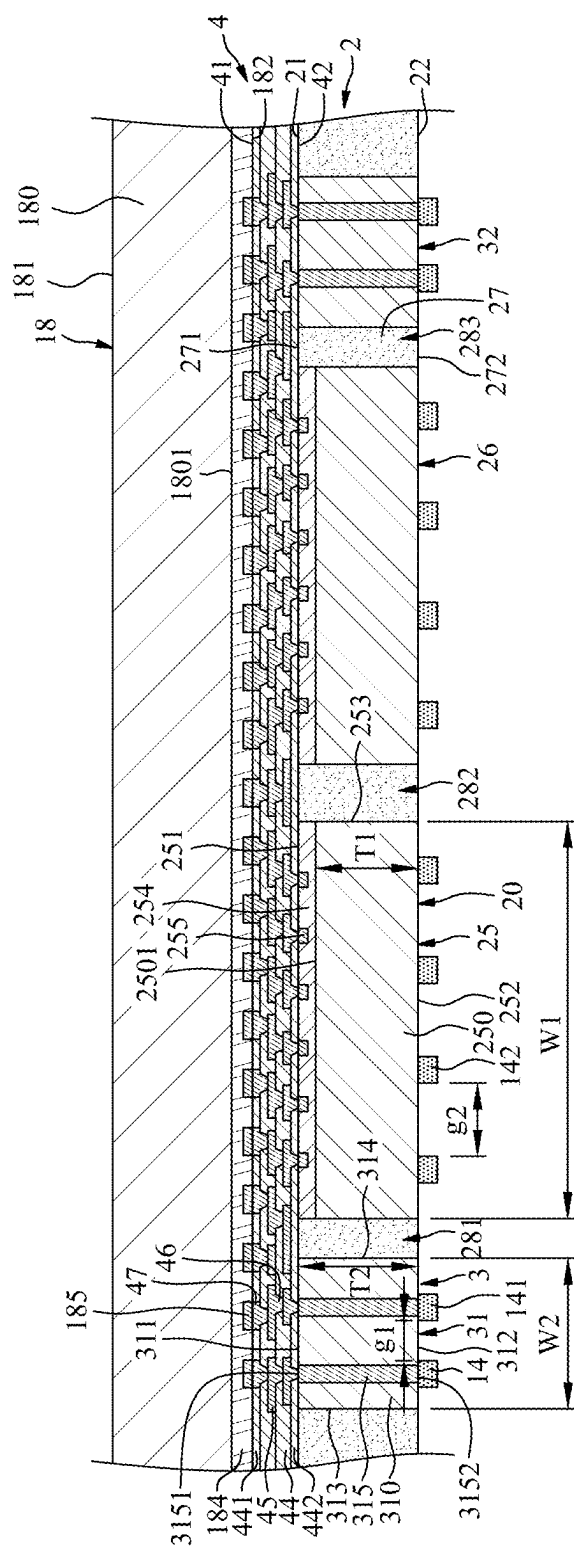


FIG. 12

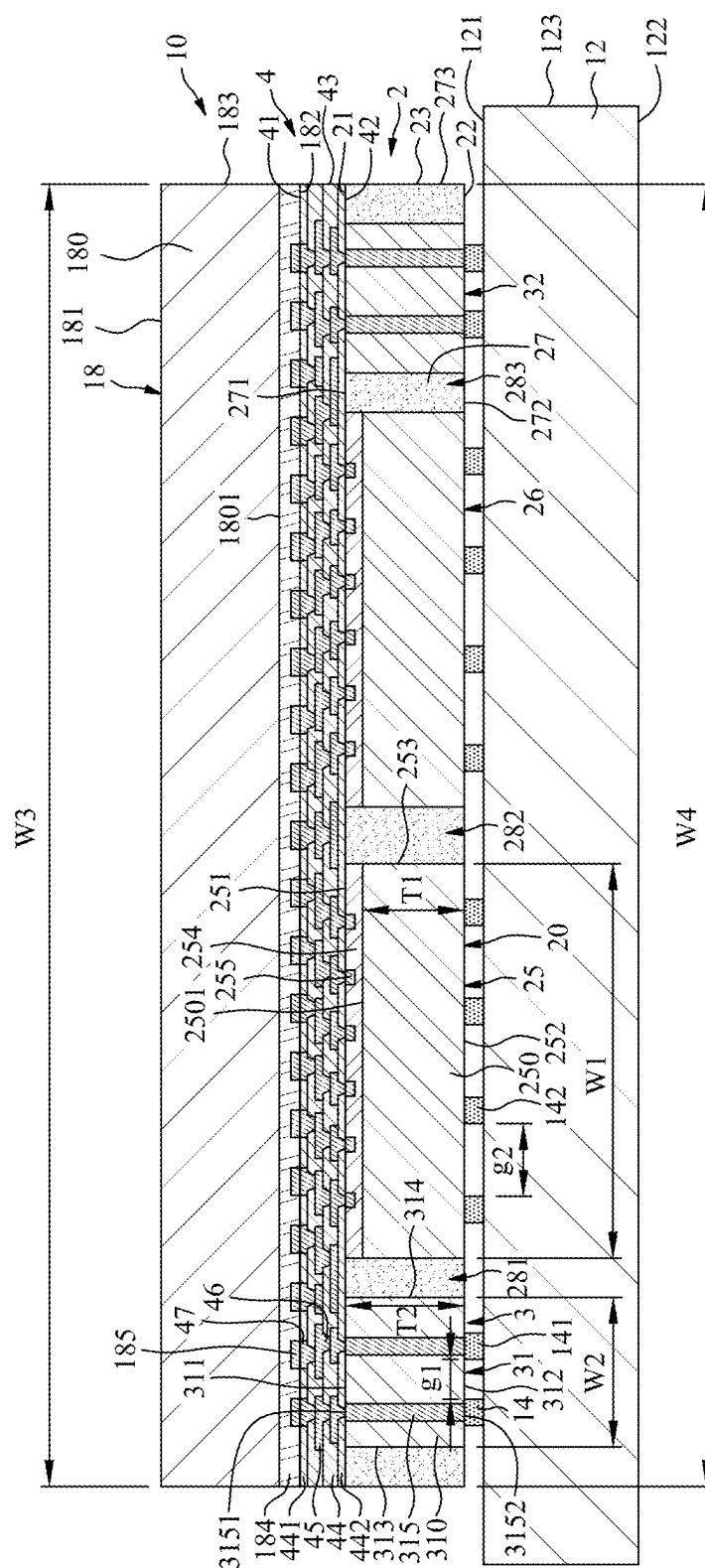


FIG. 13

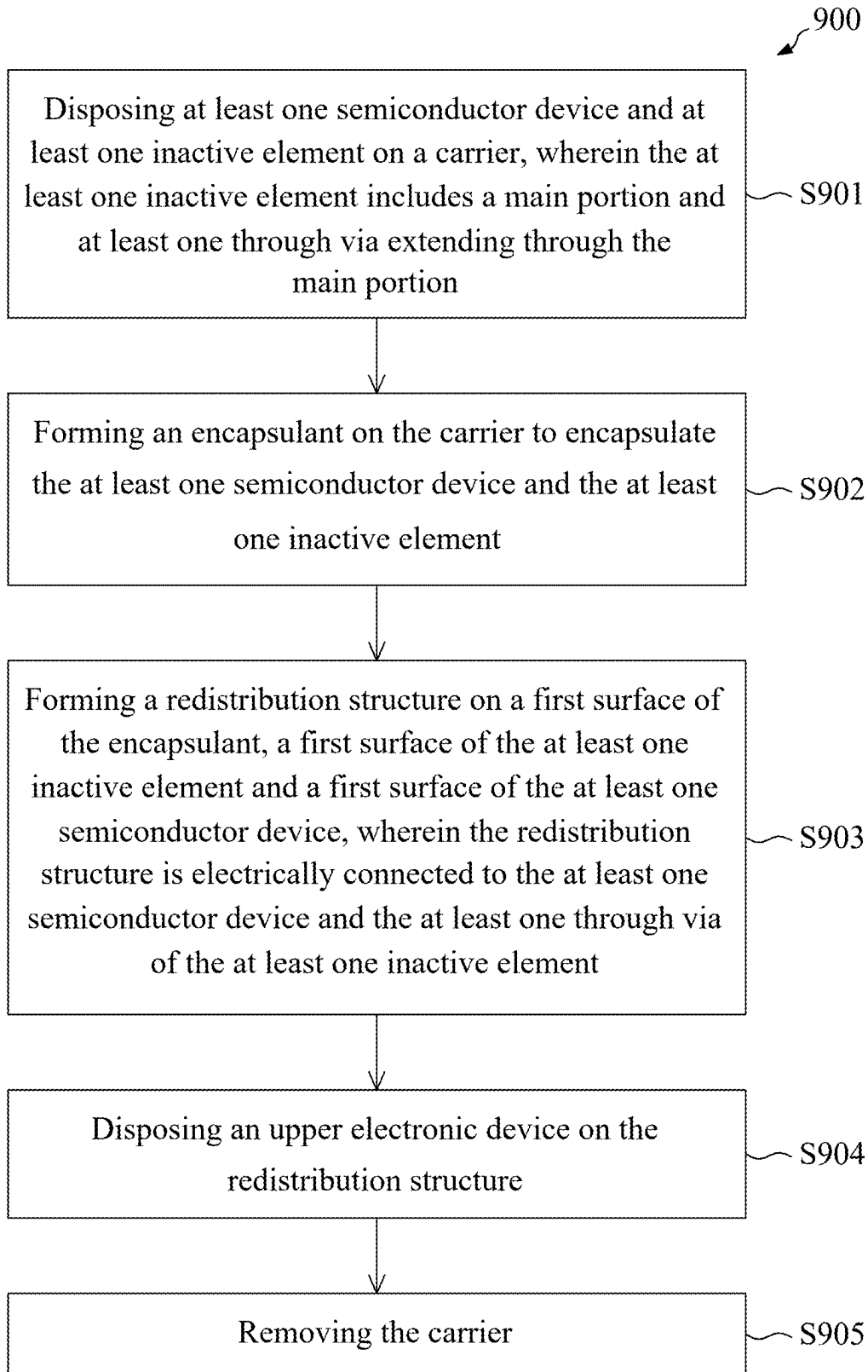


FIG. 14

**PACKAGE STRUCTURE INCLUDING
INACTIVE ELEMENT, ASSEMBLY
STRUCTURE AND METHOD OF
MANUFACTURING THE SAME**

**CROSS-REFERENCE TO RELATED
APPLICATION**

[0001] This application is a divisional application of U.S. Non-Provisional application Ser. No. 18/600,988 filed Mar. 11, 2024, which is incorporated herein by reference in its entirety.

TECHNICAL FIELD

[0002] The present disclosure relates to a package structure, an assembly structure and a method of manufacturing the same, and more particularly, to a package structure including at least one inactive element, an assembly structure including the package structure, and a method of manufacturing the same.

DISCUSSION OF THE BACKGROUND

[0003] Semiconductor electronic devices are widely used in various electronic applications, and their dimensions are constantly being reduced to meet the demands of current applications. However, scaling down semiconductor electronic devices presents several challenges that affect their final electrical characteristics, quality, cost, and yield. As semiconductor electronic devices become smaller, they require multifunctional and high-volume data processing capabilities. Consequently, there is an increasing need to enhance the integration level of semiconductor devices used in these electronic devices. However, due to the limitations of semiconductor integration technology, it is challenging to meet all the required functions with just a single semiconductor chip. To address this issue, semiconductor packages have been developed, which involve including multiple semiconductor chips.

[0004] This Discussion of the Background section is provided for background information only. The statements in this Discussion of the Background are not an admission that the subject matter disclosed herein constitutes prior art with respect to the present disclosure, and no part of this Discussion of the Background may be used as an admission that any part of this application constitutes prior art with respect to the present disclosure.

SUMMARY

[0005] One aspect of the present disclosure provides a package structure including at least one semiconductor device, at least one inactive element and an encapsulant. The at least one inactive element is disposed around the at least one semiconductor device, and includes a main portion and at least one through via extending through the main portion. The encapsulant encapsulates the at least one semiconductor device and the at least one inactive element. A first surface of the encapsulant is substantially coplanar with a first surface of the at least one inactive element and a first surface of the at least one semiconductor device. A second surface of the encapsulant is substantially coplanar with a second surface of the at least one inactive element and a second surface of the at least one semiconductor device.

[0006] Another aspect of the present disclosure provides an assembly structure including a substrate, a molded struc-

ture, a redistribution structure and an upper electronic device. The molded structure is disposed over and electrically connected to the substrate. The molded structure includes at least one semiconductor device, at least one inactive element and an encapsulant encapsulating the at least one semiconductor device and the at least one inactive element. The redistribution structure is disposed on and electrically connected to the molded structure. The upper electronic device is disposed over and electrically connected to the redistribution structure. The at least one inactive element is configured to provide a vertical conduction path between the upper electronic device and the substrate. The upper electronic device is not electrically connected to the substrate through the at least one semiconductor device.

[0007] Another aspect of the present disclosure provides a manufacturing method. The manufacturing method includes: disposing at least one semiconductor device and at least one inactive element on a carrier, wherein the at least one inactive element includes a main portion and at least one through via extending through the main portion; forming an encapsulant on the carrier to encapsulate the at least one semiconductor device and the at least one inactive element; forming a redistribution structure on a first surface of the encapsulant, a first surface of the at least one inactive element and a first surface of the at least one semiconductor device, wherein the redistribution structure is electrically connected to the at least one semiconductor device and the at least one through via of the at least one inactive element; disposing an upper electronic device on the redistribution structure; and removing the carrier.

[0008] The foregoing has outlined rather broadly the features and technical advantages of the present disclosure in order that the detailed description of the disclosure that follows may be better understood. Additional features and advantages of the disclosure will be described hereinafter, and form the subject of the claims of the disclosure. It should be appreciated by those skilled in the art that the conception and specific embodiment disclosed may be readily utilized as a basis for modifying or designing other structures or processes for carrying out the same purposes of the present disclosure. It should also be realized by those skilled in the art that such equivalent constructions do not depart from the spirit and scope of the disclosure as set forth in the appended claims.

BRIEF DESCRIPTION OF THE DRAWINGS

[0009] A more complete understanding of the present disclosure may be derived by referring to the detailed description and claims when considered in connection with the Figures, where like reference numbers refer to similar elements throughout the Figures, and:

[0010] FIG. 1 is a schematic cross-sectional view of an assembly structure in accordance with some embodiments of the present disclosure.

[0011] FIG. 2 is an enlarged view of an area “A” of FIG. 1.

[0012] FIG. 3 is a top view of the molded structure of the assembly structure of FIG. 1.

[0013] FIG. 4 is a top view of a molded structure in accordance with some embodiments of the present disclosure.

[0014] FIG. 5 is a top view of a molded structure in accordance with some embodiments of the present disclosure.

[0015] FIG. 6 is a schematic cross-sectional view of an assembly structure in accordance with some embodiments of the present disclosure.

[0016] FIG. 7 to FIG. 13 illustrate various stages of a method of manufacturing an assembly structure, in accordance with some embodiments of the present disclosure.

[0017] FIG. 14 is a flowchart of a method of manufacturing an assembly structure, in accordance with some embodiments of the present disclosure.

DETAILED DESCRIPTION

[0018] Embodiments, or examples, of the disclosure illustrated in the drawings are now described using specific language. It shall be understood that no limitation of the scope of the disclosure is hereby intended. Any alteration or modification of the described embodiments, and any further applications of principles described in this document, are to be considered as normally occurring to one of ordinary skill in the art to which the disclosure relates. Reference numerals may be repeated throughout the embodiments, but this does not necessarily mean that feature(s) of one embodiment apply to another embodiment, even if they share the same reference numeral.

[0019] It shall be understood that, although the terms first, second, third, etc., may be used herein to describe various elements, components, regions, layers or sections, these elements, components, regions, layers or sections are not limited by these terms. Rather, these terms are merely used to distinguish one element, component, region, layer or section from another region, layer or section. Thus, a first element, component, region, layer or section discussed below could be termed a second element, component, region, layer or section without departing from the teachings of the present inventive concept.

[0020] The terminology used herein is for the purpose of describing particular example embodiments only and is not intended to be limited to the present inventive concept. As used herein, the singular forms “a,” “an” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise. It shall be further understood that the terms “comprises” and “comprising,” when used in this specification, point out the presence of stated features, integers, steps, operations, elements, or components, but do not preclude the presence or addition of one or more other features, integers, steps, operations, elements, components, or groups thereof.

[0021] FIG. 1 is a schematic cross-sectional view of an assembly structure 1 in accordance with some embodiments of the present disclosure. FIG. 2 is an enlarged view of an area “A” of FIG. 1. In some embodiments, the assembly structure 1 may be a semiconductor electronic device, a semiconductor electronic structure or a package structure. In some embodiments, the assembly structure 1 may include a package structure 10, a substrate 12, a plurality of bumps 14 and a plurality of external connectors 16.

[0022] The substrate 12 may be a semiconductor substrate, and may include, for example, silicon (Si), doped silicon, germanium (Ge), silicon germanium (SiGe), silicon carbide (SiC), silicon germanium carbide (SiGeC), gallium (Ga), gallium arsenide (GaAs), indium (In), indium arsenide (InAs), indium phosphide (InP) or other IV-IV, III-V or II-VI semiconductor materials. In some embodiments, the substrate 12 may include a semiconductor-on-insulator substrate, such as a silicon-on-insulator (SOI) substrate, a

silicon germanium-on-insulator (SGOI) substrate, or a germanium-on-insulator (GOI) substrate. In some embodiments, the substrate 12 may include organic material, glass, ceramic material or the like. For example, the substrate 12 may be made of a cured photoimageable dielectric (PID) material such as epoxy or polyimide (PI) including photoinitiators. For example, the substrate 12 may include a homogeneous material. For example, the material of the substrate 12 may include epoxy type FR5, FR4, Bismaleimide triazine (BT), print circuit board (PCB) material, Prepreg (PP), Ajinomoto build-up film (ABF) or other suitable materials.

[0023] The substrate 12 may have a first surface 121 (e.g., a top surface), a second surface 122 (e.g., a bottom surface) and a lateral surface 123. The second surface 122 (e.g., the bottom surface) may be opposite to the first surface 121 (e.g., the top surface). The lateral surface 123 may extend between the first surface 121 (e.g., the top surface) and the second surface 122 (e.g., the bottom surface).

[0024] The package structure 10 may be disposed over the first surface 121 of the substrate 12, and may be attached to the first surface 121 of the substrate 12 through the bumps 14. The external connectors 16 may be disposed on the second surface 122 of the substrate 12 to provide electrical connections, for example, I/O connections, of the substrate 12. Each of the external connector 16 may include a reflowable material such as a solder ball.

[0025] The package structure 10 may include a molded structure 2, a redistribution structure 4 and an upper electronic device 18. The molded structure 2 may be disposed over and electrically connected to the substrate 12. The molded structure 2 may have a first surface 21 (e.g., a top surface), a second surface 22 (e.g., a bottom surface) and a lateral surface 23. The second surface 22 (e.g., the bottom surface) may be opposite to the first surface 21 (e.g., the top surface). The lateral surface 23 may extend between the first surface 21 (e.g., the top surface) and the second surface 22 (e.g., the bottom surface).

[0026] The molded structure 2 may include at least one semiconductor device 20, at least one inactive element 3 and an encapsulant 27. The semiconductor device 20 may include a semiconductor die or a chip, such as a memory die (e.g., dynamic random access memory (DRAM) die, static random access memory (SRAM) die, etc.). The semiconductor device 20 may include a first semiconductor device 25 and a second semiconductor device 26 disposed side by side. In some embodiments, the size and the function of the first semiconductor device 25 may be same as the size and the function of the second semiconductor device 26. The structure of the first semiconductor device 25 may be same as the structure of the second semiconductor device 26. Both of the first semiconductor device 25 and the second semiconductor device 26 may be memory dice.

[0027] The first semiconductor device 25 may have a first surface 251 (e.g., a top surface or an active surface), a second surface 252 (e.g., a bottom surface or a backside surface) and a lateral surface 253. The second surface 252 (e.g., the bottom surface) may be opposite to the first surface 251 (e.g., the top surface). The lateral surface 253 may extend between the first surface 251 (e.g., the top surface) and the second surface 252 (e.g., the bottom surface).

[0028] The first semiconductor device 25 may include a main portion 250 and an active circuit structure 254 disposed on a top surface 2501 of the main portion 250. A

material of the main portion **250** may include, for example, silicon (Si), doped silicon, germanium (Ge), silicon germanium (SiGe), silicon carbide (SiC), silicon germanium carbide (SiGeC), gallium (Ga), gallium arsenide (GaAs), indium (In), indium arsenide (InAs), indium phosphide (InP) or other IV-IV, III-V or II-VI semiconductor materials.

[0029] The active circuit structure **254** may include a plurality of dielectric layers, a plurality of circuit layers and a plurality of pads **255**. The dielectric layers may cover the circuit layers. The pads **255** may be electrically connected to the circuit layers, and embedded in the dielectric layers. The pads **255** may be exposed from the first surface **251** of the first semiconductor device **25**.

[0030] The inactive element **3** may be disposed around the semiconductor device **20** (including, e.g., the first semiconductor device **25** and the second semiconductor device **26**). The inactive element **3** may be also referred to as “a dummy die”. The inactive element **3** may include a first inactive element **31** and a second inactive element **32** disposed around the first semiconductor device **25** and the second semiconductor device **26**. A structure of the first inactive element **31** may be same as or similar to a structure of the second inactive element **32**.

[0031] The first inactive element **31** may have a first surface **311** (e.g., a top surface), a second surface **312** (e.g., a bottom surface) and two lateral surfaces **313**, **314**. The second surface **312** (e.g., the bottom surface) may be opposite to the first surface **311** (e.g., the top surface). The lateral surfaces **313**, **314** may extend between the first surface **311** (e.g., the top surface) and the second surface **312** (e.g., the bottom surface).

[0032] The first inactive element **31** may include a main portion **310** and at least one through via **315** (or through silicon via (TSV)) extending through the main portion **310**. The main portion **310** may have a first surface **311** (e.g., a top surface), a second surface **312** (e.g., a bottom surface) and two lateral surfaces **313**, **314**. The second surface **312** (e.g., the bottom surface) may be opposite to the first surface **311** (e.g., the top surface). The lateral surfaces **313**, **314** may extend between the first surface **311** (e.g., the top surface) and the second surface **312** (e.g., the bottom surface). Thus, the first surface **311**, the second surface **312** and the lateral surfaces **313**, **314** of the first inactive element **31** may be the first surface **311**, the second surface **312** and the lateral surfaces **313**, **314** of the main portion **310**, respectively.

[0033] A material of the main portion **310** may include, for example, silicon (Si), doped silicon, germanium (Ge), silicon germanium (SiGe), silicon carbide (SiC), silicon germanium carbide (SiGeC), gallium (Ga), gallium arsenide (GaAs), indium (In), indium arsenide (InAs), indium phosphide (InP) or other IV-IV, III-V or II-VI semiconductor materials. In some embodiments, the material of the main portion **250** of the first semiconductor device **25** may be same as the material of the main portion **310** of the first inactive element **31**.

[0034] A width **W1** of the first semiconductor device **25** (or the main portion **250**) may be greater than a width **W2** of the first inactive element **31** (or the main portion **310**). A thickness **T1** of the main portion **250** of the first semiconductor device **25** may be less than a thickness **T2** of the main portion **310** of the first inactive element **31**.

[0035] The semiconductor device **20** (including, e.g., the first semiconductor device **25** and the second semiconductor device **26**) does not include a vertical conduction path in the

main portion thereof. That is, the semiconductor device **20** (including, e.g., the first semiconductor device **25** and the second semiconductor device **26**) may be free of a vertical conduction path to the substrate **12**. For example, the first semiconductor device **25** does not include a vertical conduction path in the main portion **250** thereof.

[0036] The through via **315** of the first inactive element **31** may have a first surface **3151** (e.g., a top surface) and a second surface **3152** (e.g., a bottom surface) opposite to the first surface **3151** (e.g., the top surface). The first surface **3151** of the first through via **315** of the first inactive element **31** may be substantially coplanar with and exposed from the first surface **311** of the first inactive element **31**. The second surface **3152** of the through via **315** of the first inactive element **31** may be substantially coplanar with and exposed from the second surface **312** of the first inactive element **31**. That is, the through via **315** of the first inactive element **31** may extend from the first surface **311** of the main portion **310** to the second surface **312** of the main portion **310**.

[0037] Thus, the through via **315** of the first inactive element **31** may provide a vertical conduction path extending through the main portion **310** of the first inactive element **31**, and may be configured to transmit signals or power. Thus, the redistribution structure **4** and the upper electronic device **18** may be electrically connected to the substrate **12** through the through via **315** of the first inactive element **31** rather than through the semiconductor device **20** (including, e.g., the first semiconductor device **25** and the second semiconductor device **26**). There is no need to form a through via in the main portion (e.g., the main portion **250**) of the semiconductor device **20** (e.g., the first semiconductor device **25** and the second semiconductor device **26**).

[0038] In some embodiments, the inactive element **3** does not include a horizontal conduction path (e.g., a horizontal circuit layer) on the main portion thereof. For example, the first inactive element **31** does not include a horizontal conduction path (e.g., a horizontal circuit layer) on the main portion **310** thereof. There is no horizontal conduction path on the first surface **311** of the main portion **310** nor on the second surface **312** of the main portion **310**. The inactive element **3** only include the vertical conduction path.

[0039] The encapsulant **27** may encapsulate the semiconductor device **20** (including, e.g., the first semiconductor device **25** and the second semiconductor device **26**) and the inactive element **3** (including, e.g., the first inactive element **31** and the second inactive element **32**). For example, the encapsulant **27** may be disposed in a space **281** between the first semiconductor device **25** and the first inactive element **31**, a space **282** between the first semiconductor device **25** and the second semiconductor device **26**, and a space **283** between the second semiconductor device **26** and the second inactive element **32**. A material of the encapsulant **27** may include a molding compound with or without fillers.

[0040] The encapsulant **27** may have a first surface **271** (e.g., a top surface), a second surface **272** (e.g., a bottom surface) and a lateral surface **273**. The second surface **272** (e.g., the bottom surface) may be opposite to the first surface **271** (e.g., the top surface). The lateral surface **273** may extend between the first surface **271** (e.g., the top surface) and the second surface **272** (e.g., the bottom surface).

[0041] The first surface **271** of the encapsulant **27** may be substantially coplanar with or aligned with the first surface **311** of the first inactive element **31** and the first surface **251** of the first semiconductor device **25**. Thus, the first surface

21 of the molded structure **2** may include the first surface **271** of the encapsulant **27**, the first surface **311** of the first inactive element **31** and the first surface **251** of the first semiconductor device **25**. In addition, the second surface **272** of the encapsulant **27** may be substantially coplanar with or aligned with the second surface **312** of the first inactive element **31** and the second surface **252** of the first semiconductor device **25**. Thus, the second surface **22** of the molded structure **2** may include the second surface **272** of the encapsulant **27**, the second surface **312** of the first inactive element **31** and the second surface **252** of the first semiconductor device **25**.

[0042] The redistribution structure **4** may be disposed on and electrically connected to the first surface **21** of the molded structure **2**. For example, the redistribution structure **4** may be disposed on the first surface **271** of the encapsulant **27**, the first surface **311** of the first inactive element **31** and the first surface **251** of the first semiconductor device **25**, and electrically connected to the first semiconductor device **25** and the through via **315** of the first inactive element **31**.

[0043] The redistribution structure **4** may be a fan-out structure. The redistribution structure **4** may have a first surface **41** (e.g., a top surface), a second surface **42** (e.g., a bottom surface) and a lateral surface **43**. The second surface **42** (e.g., the bottom surface) may be opposite to the first surface **41** (e.g., the top surface). The lateral surface **43** may extend between the first surface **41** (e.g., the top surface) and the second surface **42** (e.g., the bottom surface). The second surface **42** of the redistribution structure **4** may contact the first surface **21** of the molded structure **2** directly.

[0044] The redistribution structure **4** may include a plurality of dielectric layers **44**, a plurality of circuit layers **45**, a plurality of inner vias **46** and a plurality of pads **47**. The dielectric layers **44** may be made of a cured photoimageable dielectric (PID) material such as epoxy or polyimide (PI) including photoinitiators. The material of the dielectric layers **44** may be the same with each other. In some embodiments, the material of the topmost dielectric layer **441** may be different from the material of the other dielectric layers **44**. The topmost dielectric layer **441** may be a hybrid bonding (HB) dielectric layer, and may include SiO₂, SiCN and/or SiON. In some embodiments, the bottommost dielectric layer **442** may be omitted. The circuit layers **45** may be disposed on the first surface **21** of the molded structure **2** directly.

[0045] The circuit layers **45** may be covered by or embedded in the dielectric layers **44**. The circuit layers **45** may be fan-out circuit layers. The inner vias **46** may be embedded in the dielectric layers **44**, and may connect adjacent two of the circuit layers **45**. The inner vias **46** may taper toward the molded structure **2**. The pads **47** may be electrically connected to the circuit layers **45**, and embedded in the dielectric layers **44** (e.g., the topmost dielectric layer **441**). The pads **47** may be exposed from the first surface **41** of the redistribution structure **4**. Each of the pads **47** may be a hybrid bonding (HB) pad, and may include Cu or Al.

[0046] The upper electronic device **18** may be disposed over and electrically connected to the first surface **41** of the redistribution structure **4**. The upper electronic device **18** may be also disposed over and electrically connected to the first surface **21** of the molded structure **2** (including, e.g., the first surface **271** of the encapsulant **27**, the first surface **311** of the first inactive element **31** and the first surface **251** of the first semiconductor device **25**).

[0047] The upper electronic device **18** may include a semiconductor die or a chip, such as a signal processing die (e.g., digital signal processing (DSP) die), a logic die (e.g., application processor (AP), system-on-a-chip (SoC), central processing unit (CPU), graphics processing unit (GPU), microcontroller, etc.), a radio frequency (RF) die, a sensor die, a micro-electro-mechanical-system (MEMS) die, a front-end die (e.g., analog front-end (AFE) dies) or other active components.

[0048] The upper electronic device **18** may have a first surface **181** (e.g., a top surface or a backside surface), a second surface **182** (e.g., a bottom surface or an active surface) and a lateral surface **183**. The second surface **182** (e.g., the bottom surface) may be opposite to the first surface **181** (e.g., the top surface). The lateral surface **183** may extend between the first surface **181** (e.g., the top surface) and the second surface **182** (e.g., the bottom surface).

[0049] The upper electronic device **18** may include a main portion **180** and an active circuit structure **184** disposed on a bottom surface **1801** of the main portion **180**. A material of the main portion **180** may include, for example, silicon (Si), doped silicon, germanium (Ge), silicon germanium (SiGe), silicon carbide (SiC), silicon germanium carbide (SiGeC), gallium (Ga), gallium arsenide (GaAs), indium (In), indium arsenide (InAs), indium phosphide (InP) or other IV-IV, III-V or II-VI semiconductor materials.

[0050] The active circuit structure **184** may include a plurality of dielectric layers, a plurality of circuit layers and a plurality of pads **185**. The dielectric layers may cover the circuit layers. The pads **185** may be electrically connected to the circuit layers, and embedded in the dielectric layers. The pads **185** may be exposed from the second surface **182** of the upper electronic device **18**. The active circuit structure **184** may include a bottommost dielectric layer surrounding the pads **185**. The bottommost dielectric layer may be a hybrid bonding (HB) dielectric layer, and may include SiO₂, SiCN and/or SiON. Each of the pads **185** may be a hybrid bonding (HB) pad, and may include Cu or Al.

[0051] The upper electronic device **18** may be attached to and electrically connected to the first surface **41** of the redistribution structure **4** by hybrid bonding. That is, the second surface **182** of the upper electronic device **18** may contact the first surface **41** of the redistribution structure **4** directly. Thus, the active circuit structure **184** of the upper electronic device **18** may be attached to, may connect to and may contact the topmost dielectric layer **441** of the redistribution structure **4** directly. The pads **185** of the upper electronic device **18** may be attached to, may connect to and may contact the pads **47** of the redistribution structure **4** directly.

[0052] The size and the function of the upper electronic device **18** may be different from the size and the function of the semiconductor device **20** (including, e.g., the first semiconductor device **25** and the second semiconductor device **26**). A width **W3** of the upper electronic device **18** may be substantially equal to a width **W4** of the molded structure **2** or a width **W4** of the encapsulant **27**. The lateral surface **183** of the upper electronic device **18** may be substantially aligned with the lateral surface **23** of the molded structure **2** or the lateral surface **273** of the encapsulant **27**. The upper electronic device **18** may vertically overlap the inactive element **3** (including, e.g., the first inactive element **31** and the second inactive element **32**).

[0053] The bumps 14 may be disposed on the second surface 22 of the molded structure 2. For example, the bumps 14 may include a plurality of first bumps 141 and a plurality of second bumps 142. The first bumps 141 may be disposed on the second surface 3152 (e.g., the bottom surface) of the through via 315. Thus, the first bumps 141 may physically connect and electrically connect the through via 315 and the first surface 121 of the substrate 12. The second bumps 142 may be disposed on the second surface of the semiconductor device 20 (e.g., the second surface 252 of the first semiconductor device 25). Thus, second bumps 142 may connect the second surface of the semiconductor device 20 (e.g., the second surface 252 of the first semiconductor device 25) and the first surface 121 of the substrate 12. The second bumps 142 may have no electrical function, and may be dummy pads.

[0054] In some embodiments, the first bumps 141 and the second bumps 142 may be formed concurrently at a manufacturing stage. In some embodiments, the first bumps 141 and/or the second bumps 142 may include a reflowable material configured to control a gap between the molded structure 2 and the substrate 12 so as to prevent the molded structure 2 from tilting with respect to the substrate 12. In addition, a first gap g1 between adjacent two of the first bumps 141 may be less than a second gap g2 between adjacent two of the second bumps 142.

[0055] FIG. 3 is a top view of the molded structure 2 of the assembly structure 1 of FIG. 1. The first inactive element 31 may include a plurality of rows of through vias 315, for example, two rows of through vias 315. The pads 255 of the first semiconductor device 25 may be arranged in an array. The size and the structure of the second semiconductor device 26 may be same as the size and the structure of the first semiconductor device 25. The size and the structure of the second inactive element 32 may be same as the size and the structure of the first inactive element 31. The second inactive element 32 and the first inactive element 31 may be disposed on opposite sides of the second semiconductor device 26 and the first semiconductor device 25.

[0056] In the embodiment illustrated in FIG. 1 to FIG. 3, the inactive element 3 (including, e.g., the first inactive element 31 and the second inactive element 32) is configured to provide a vertical conduction path between the upper electronic device 18 and the substrate 12. The upper electronic device 18 is not electrically connected to the substrate 12 through the semiconductor device 20 (e.g., the first semiconductor device 25 and the second semiconductor device 26). It is not necessary to form a through via in the main portion (e.g., the main portion 250) of the semiconductor device 20 (e.g., the first semiconductor device 25 and the second semiconductor device 26). The vertical conduction path may be disposed outside the semiconductor device 20 (e.g., the first semiconductor device 25 and the second semiconductor device 26). Thus, the design flexibility is increased, and the manufacturing cost is reduced.

[0057] In addition, the active surface of the semiconductor device 20 (e.g., the first surface 251 (or the active surface) of the first semiconductor device 25) faces the active surface of the upper electronic device 18 (e.g., the first surface 181 of the upper electronic device 18), thus, the upper electronic device 18 can read from and/or write to the semiconductor device 20 (e.g., the first semiconductor device 25) quickly. Thus, the performance the package structure 10 can be improved.

[0058] FIG. 4 is a top view of a molded structure 2a in accordance with some embodiments of the present disclosure. The molded structure 2a may be similar to the molded structure 2 of FIG. 3, and the differences are described as follows. The first inactive element 31 and the second inactive element 32 may include three rows of through vias 315. The molded structure 2a may further include a third first inactive element 33 and a fourth inactive element 34. The third first inactive element 33 and the fourth inactive element 34 may be disposed on opposite sides of the second semiconductor device 26 and the first semiconductor device 25. The size and the structure of the third first inactive element 33 and the fourth inactive element 34 may be same as or similar to the size and the structure of the first inactive element 31 and the second inactive element 32. The first inactive element 31, the second inactive element 32, the third first inactive element 33 and the fourth inactive element 34 may surround the second semiconductor device 26 and the first semiconductor device 25.

[0059] FIG. 5 is a top view of a molded structure 2b in accordance with some embodiments of the present disclosure. The molded structure 2b may be similar to the molded structure 2a of FIG. 4, and the differences are described as follows. A size of the first inactive element 31 may be different from and a size of the second inactive element 32b from the top view. For example, a length L1 of the first inactive element 31 may be greater than a length L2 of the second inactive element 32b. In addition, each of the third first inactive element 33b and the fourth inactive element 34b may be in an L shape, and may be disposed adjacent to or disposed around a corner of the second semiconductor device 26.

[0060] FIG. 6 is a schematic cross-sectional view of an assembly structure 1a in accordance with some embodiments of the present disclosure. The assembly structure 1a may be similar to the assembly structure 1 of FIG. 1 except for the structure of the package structure 10a. In the package structure 10a, the second surface 182 of the upper electronic device 18 does not contact the first surface 41 of the redistribution structure 4. The second surface 182 of the upper electronic device 18 is spaced apart from the first surface 41 of the redistribution structure 4. The pads 185 of the upper electronic device 18 may be electrically connected to the pads 47 of the redistribution structure 4 through a plurality of solder materials 19. The solder materials 19 may include a reflowable material such as AgSn. Thus, the upper electronic device 18 is electrically connected to the redistribution structure 4 through solder bonding instead of hybrid bonding. The pads 185 of the upper electronic device 18 and the pads 47 of the redistribution structure 4 may be solder bonding pads.

[0061] FIG. 7 to FIG. 13 illustrate various stages of a method of manufacturing an assembly structure 1, in accordance with some embodiments of the present disclosure.

[0062] Referring to FIG. 7, at least one semiconductor device 20 and at least one inactive element 3 may be disposed side by side on a carrier 80. In some embodiments, the carrier 80 may include a release layer 82 on a surface thereof. The semiconductor device 20 and the inactive element 3 may be disposed on the release layer 82. In some embodiments, the singulated semiconductor device 20 (e.g., the first semiconductor device 25 and the second semiconductor device 26) and the singulated inactive element 3 (e.g., the first inactive element 31 and the second inactive element

32) may be reconstructed or rearranged on the release layer 82 of the carrier 80. In some embodiments, only known good dice, e.g., the known good semiconductor device 20 (e.g., the first semiconductor device 25 and the second semiconductor device 26) and the known good inactive element 3 (e.g., the first inactive element 31 and the second inactive element 32) are used.

[0063] In some embodiments, the size and the function of the first semiconductor device 25 may be same as the size and the function of the second semiconductor device 26. Both of the first semiconductor device 25 and the second semiconductor device 26 may be memory dice.

[0064] The first semiconductor device 25 may have a first surface 251 (e.g., a top surface or an active surface), a second surface 252 (e.g., a bottom surface or a backside surface) and a lateral surface 253. The second surface 252 (e.g., the bottom surface) may be opposite to the first surface 251 (e.g., the top surface). The lateral surface 253 may extend between the first surface 251 (e.g., the top surface) and the second surface 252 (e.g., the bottom surface).

[0065] The first semiconductor device 25 may include a main portion 250 and an active circuit structure 254 disposed on a top surface 2501 of the main portion 250. The active circuit structure 254 may include a plurality of dielectric layers, a plurality of circuit layers and a plurality of pads 255. The dielectric layers may cover the circuit layers. The pads 255 may be electrically connected to the circuit layers, and embedded in the dielectric layers. The pads 255 may be exposed from the first surface 251 of the first semiconductor device 25. The active circuit structure 254 and the pads 255 may face up. The second surface 252 (e.g., the bottom surface) of the first semiconductor device 25 may contact the release layer 82 of the carrier 80.

[0066] The inactive element 3 may include a first inactive element 31 and a second inactive element 32 disposed around the first semiconductor device 25 and the second semiconductor device 26. A structure of the first inactive element 31 may be same as or similar to a structure of the second inactive element 32.

[0067] The first inactive element 31 may have a first surface 311 (e.g., a top surface), a second surface 312 (e.g., a bottom surface) and two lateral surfaces 313, 314. The second surface 312 (e.g., the bottom surface) may be opposite to the first surface 311 (e.g., the top surface). The lateral surfaces 313, 314 may extend between the first surface 311 (e.g., the top surface) and the second surface 312 (e.g., the bottom surface).

[0068] A space 281 may be formed between the first semiconductor device 25 and the first inactive element 31. A space 282 may be formed between the first semiconductor device 25 and the second semiconductor device 26. A space 283 may be formed between the second semiconductor device 26 and the second inactive element 32.

[0069] The first inactive element 31 may include a main portion 310 and at least one through via 315 (or through silicon via (TSV)) extending through the main portion 310. The main portion 310 may have a first surface 311 (e.g., a top surface), a second surface 312 (e.g., a bottom surface) and two lateral surfaces 313, 314. The second surface 312 (e.g., the bottom surface) may be opposite to the first surface 311 (e.g., the top surface). The lateral surfaces 313, 314 may extend between the first surface 311 (e.g., the top surface) and the second surface 312 (e.g., the bottom surface). Thus, the first surface 311, the second surface 312 and the lateral

surfaces 313, 314 of the first inactive element 31 may be the first surface 311, the second surface 312 and the lateral surfaces 313, 314 of the main portion 310, respectively.

[0070] In some embodiments, the material of the main portion 250 of the first semiconductor device 25 may be same as the material of the main portion 310 of the first inactive element 31.

[0071] A width W1 of the first semiconductor device 25 (or the main portion 250) may be greater than a width W2 of the first inactive element 31 (or the main portion 310). A thickness T1 of the main portion 250 of the first semiconductor device 25 may be less than a thickness T2 of the main portion 310 of the first inactive element 31.

[0072] The semiconductor device 20 (including, e.g., the first semiconductor device 25 and the second semiconductor device 26) does not include a vertical conduction path in the main portion thereof. For example, the first semiconductor device 25 does not include a vertical conduction path in the main portion 250 thereof.

[0073] The through via 315 of the first inactive element 31 may have a first surface 3151 (e.g., a top surface) and a second surface 3152 (e.g., a bottom surface) opposite to the first surface 3151 (e.g., the top surface). The first surface 3151 of the first through via 315 of the first inactive element 31 may be substantially coplanar with and exposed from the first surface 311 of the first inactive element 31. The second surface 3152 of the through via 315 of the first inactive element 31 may be substantially coplanar with and exposed from the second surface 312 of the first inactive element 31. That is, the through via 315 of the first inactive element 31 may extend from the first surface 311 of the main portion 310 to the second surface 312 of the main portion 310.

[0074] In some embodiments, the inactive element 3 does not include a horizontal conduction path (e.g., a horizontal circuit layer) on the main portion thereof. For example, the first inactive element 31 does not include a horizontal conduction path (e.g., a horizontal circuit layer) on the main portion 310 thereof. There is no horizontal conduction path on the first surface 311 of the main portion 310 nor on the second surface 312 of the main portion 310. The inactive element 3 only include the vertical conduction path. The second surface 312 of the inactive element 3 may contact the release layer 82 of the carrier 80.

[0075] Referring to FIG. 8, an encapsulant 27 may be formed on the release layer 82 of the carrier 80 to encapsulate the semiconductor device 20 (including, e.g., the first semiconductor device 25 and the second semiconductor device 26) and the inactive element 3 (e.g., the first inactive element 31 and the second inactive element 32). The encapsulant 27 may be disposed in the space 281, the space 282 and the space 283.

[0076] Then, a grinding process may be conducted to form a molded structure 2 on the carrier 80. The molded structure 2 may include the semiconductor device 20, the inactive element 3 and the encapsulant 27. The molded structure 2 may have a first surface 21 (e.g., a top surface) and a second surface 22 (e.g., a bottom surface) opposite to the first surface 21 (e.g., the top surface).

[0077] The encapsulant 27 may have a first surface 271 (e.g., a top surface) and a second surface 272 (e.g., a bottom surface) opposite to the first surface 271 (e.g., the top surface). The first surface 271 of the encapsulant 27 may be substantially coplanar with or aligned with the first surface 311 of the first inactive element 31 and the first surface 251

of the first semiconductor device 25. Thus, the first surface 21 of the molded structure 2 may include the first surface 271 of the encapsulant 27, the first surface 311 of the first inactive element 31 and the first surface 251 of the first semiconductor device 25. In addition, the second surface 272 of the encapsulant 27 may be substantially coplanar with or aligned with the second surface 312 of the first inactive element 31 and the second surface 252 of the first semiconductor device 25. Thus, the second surface 22 of the molded structure 2 may include the second surface 272 of the encapsulant 27, the second surface 312 of the first inactive element 31 and the second surface 252 of the first semiconductor device 25.

[0078] Referring to FIG. 9, a redistribution structure 4 may be formed or disposed on the first surface 21 of the molded structure 2. For example, the redistribution structure 4 may be formed or disposed on the first surface 271 of the encapsulant 27, the first surface 311 of the first inactive element 31 and the first surface 251 of the first semiconductor device 25. The redistribution structure 4 may be electrically connected to the semiconductor device 20 (e.g., the first semiconductor device 25 and the second semiconductor device 26) and through via 315 of the inactive element 3 (e.g., the first inactive element 31 and the second inactive element 32).

[0079] The redistribution structure 4 may be a fan-out structure. The redistribution structure 4 may have a first surface 41 (e.g., a top surface) and a second surface 42 (e.g., a bottom surface) opposite to the first surface 41 (e.g., the top surface). The second surface 42 of the redistribution structure 4 may contact the first surface 21 of the molded structure 2 directly.

[0080] The redistribution structure 4 may include a plurality of dielectric layers 44, a plurality of circuit layers 45, a plurality of inner vias 46 and a plurality of pads 47. The material of the dielectric layers 44 may be the same with each other. In some embodiments, the material of the topmost dielectric layer 441 may be different from the material of the other dielectric layers 44. The topmost dielectric layer 441 may be a hybrid bonding (HB) dielectric layer. In some embodiments, the bottommost dielectric layer 442 may be omitted. The circuit layers 45 may be disposed on the first surface 21 of the molded structure 2 directly.

[0081] The circuit layers 45 may be covered by or embedded in the dielectric layers 44. The circuit layers 45 may be fan-out circuit layers. The inner vias 46 may be embedded in the dielectric layers 44, and may connect adjacent two of the circuit layers 45. The inner vias 46 may taper toward the molded structure 2. The pads 47 may be electrically connected to the circuit layers 45, and embedded in the dielectric layers 44 (e.g., the topmost dielectric layer 441). The pads 47 may be exposed from the first surface 41 of the redistribution structure 4. Each of the pads 47 may be a hybrid bonding (HB) pad.

[0082] Referring to FIG. 10, an upper electronic device 18 may be disposed on and electrically connected to the first surface 41 of the redistribution structure 4. The upper electronic device 18 may be in a wafer type, a panel type or a chip type. The upper electronic device 18 may have a first surface 181 (e.g., a top surface or a backside surface) and a second surface 182 (e.g., a bottom surface or an active surface) opposite to the first surface 181 (e.g., the top surface).

[0083] The upper electronic device 18 may include a main portion 180 and an active circuit structure 184 disposed on a bottom surface 1801 of the main portion 180. The active circuit structure 184 may include a plurality of dielectric layers, a plurality of circuit layers and a plurality of pads 185. The dielectric layers may cover the circuit layers. The pads 185 may be electrically connected to the circuit layers, and embedded in the dielectric layers. The pads 185 may be exposed from the second surface 182 of the upper electronic device 18. The active circuit structure 184 may include a bottommost dielectric layer surrounding the pads 185. The bottommost dielectric layer may be a hybrid bonding (HB) dielectric layer. Each of the pads 185 may be a hybrid bonding (HB) pad.

[0084] The upper electronic device 18 may be attached to and electrically connected to the first surface 41 of the redistribution structure 4 by hybrid bonding. That is, the second surface 182 of the upper electronic device 18 may contact the first surface 41 of the redistribution structure 4 directly. Thus, the active circuit structure 184 of the upper electronic device 18 may be attached to, may connect to and may contact the topmost dielectric layer 441 of the redistribution structure 4 directly. The pads 185 of the upper electronic device 18 may be attached to, may connect to and may contact the pads 47 of the redistribution structure 4 directly.

[0085] Referring to FIG. 11, the release layer 82 and the carrier 80 may be removed from the molded structure 2.

[0086] Referring to FIG. 12, a plurality of bumps 14 may be formed or disposed on the second surface 22 of the molded structure 2. For example, the bumps 14 may include a plurality of first bumps 141 and a plurality of second bumps 142. The first bumps 141 may be formed or disposed on the second surface 3152 (e.g., the bottom surface) of the through via 315. The second bumps 142 may be formed or disposed on the second surface of the semiconductor device 20 (e.g., the second surface 252 of the first semiconductor device 25).

[0087] In some embodiments, the first bumps 141 and the second bumps 142 may be formed concurrently at a same stage. In some embodiments, the first bumps 141 and/or the second bumps 142 may include a reflowable material. In addition, a first gap g1 between adjacent two first bumps 141 may be less than a second gap g2 between adjacent two second bumps 142. Alternatively, the first gap g1 may be equal to the second gap g2.

[0088] Referring to FIG. 13, the encapsulant 27, the redistribution structure 4 and the upper electronic device 18 may be diced to form a package structure 10. Thus, a lateral surface 183 of the upper electronic device 18 may be substantially aligned with a lateral surface 23 of the molded structure 2 (or a lateral surface 273 of the encapsulant 27) and a lateral surface 43 of the redistribution structure 4. A width W3 of the upper electronic device 18 may be substantially equal to a width W4 of the molded structure 2 or a width W4 of the encapsulant 27.

[0089] Then, the bumps 14 (e.g., the first bumps 141 and the second bumps) may be attached to a substrate 12. Thus, the package structure 10 may be attached to the substrate 12 through the bumps 14 (e.g., the first bumps 141 and the second bumps).

[0090] Then, a plurality of external connectors 16 may be formed or disposed on a second surface 122 of the substrate

12 to provide electrical connections, for example, I/O connections, of the substrate 12. Therefore, the assembly structure 1 of FIG. 1 is obtained.

[0091] FIG. 14 illustrates a flow chart of a method 900 of manufacturing an assembly structure 1 in accordance with some embodiments of the present disclosure.

[0092] In some embodiments, the method 900 may include a step S901, disposing at least one semiconductor device and at least one inactive element on a carrier, wherein the at least one inactive element includes a main portion and at least one through via extending through the main portion. For example, as shown in FIG. 7, at least one semiconductor device 20 (e.g., the second surface 252 of the first semiconductor device 25) and at least one inactive element 3 (e.g., the first inactive element 31 and the second inactive element 32) may be disposed on a carrier 80. The inactive element 31 may include a main portion 310 and at least one through via 315 extending through the main portion 310.

[0093] In some embodiments, the method 900 may include a step S902, forming an encapsulant on the carrier to encapsulate the at least one semiconductor device and the at least one inactive element. For example, as shown in FIG. 8, an encapsulant 27 may be formed on the carrier 80 to encapsulate the at least one semiconductor device 20 (e.g., the second surface 252 of the first semiconductor device 25) and the at least one inactive element 3 (e.g., the first inactive element 31 and the second inactive element 32).

[0094] In some embodiments, the method 900 may include a step S903, forming a redistribution structure on a first surface of the encapsulant, a first surface of the at least one inactive element and a first surface of the at least one semiconductor device, wherein the redistribution structure is electrically connected to the at least one semiconductor device and the at least one through via of the at least one inactive element. For example, as shown in FIG. 9, a redistribution structure 4 may be formed on a first surface 271 of the encapsulant 27, a first surface 311 of the at least one inactive element 31 and a first surface 251 of the at least one semiconductor device 25. The redistribution structure 4 may be electrically connected to the at least one semiconductor device 25 and the at least one through via 315 of the at least one inactive element 31.

[0095] In some embodiments, the method 900 may include a step S904, disposing an upper electronic device on the redistribution structure. For example, as shown in FIG. 10, an upper electronic device 18 may be disposed on the redistribution structure 4.

[0096] In some embodiments, the method 900 may include a step S905, disposing an upper electronic device on the redistribution structure. For example, as shown in FIG. 11, the carrier 80 may be removed.

[0097] One aspect of the present disclosure provides an electronic device including a first semiconductor chip, a second semiconductor chip and a third semiconductor chip. The second semiconductor chip is stacked on the first semiconductor chip, and is electrically connected to the first semiconductor chip by hybrid bonding. The third semiconductor chip is stacked on the second semiconductor chip, and is electrically connected to the second semiconductor chip through a plurality of bumps.

[0098] Another aspect of the present disclosure provides an electronic device including a first assembly and a second assembly. The first assembly includes a first semiconductor chip and a second semiconductor chip stacked on the first

semiconductor chip and electrically connected to the first semiconductor chip by hybrid bonding. The second assembly includes a third semiconductor chip and a fourth semiconductor chip stacked on the third semiconductor chip and electrically connected to the third semiconductor chip by hybrid bonding. The second assembly is electrically connected to the first assembly through a plurality of bumps.

[0099] Another aspect of the present disclosure provides a manufacturing method. The manufacturing method includes providing a first assembly including a first semiconductor chip and a second semiconductor chip stacked on the first semiconductor chip and electrically connected to the first semiconductor chip by hybrid bonding; providing a second assembly including a third semiconductor chip and a fourth semiconductor chip stacked on the third semiconductor chip and electrically connected to the third semiconductor chip by hybrid bonding; and electrically connecting the second assembly to the first assembly through a plurality of bumps.

[0100] Although the present disclosure and its advantages have been described in detail, it should be understood that various changes, substitutions and alterations can be made herein without departing from the spirit and scope of the disclosure as defined by the appended claims. For example, many of the processes discussed above can be implemented in different methodologies and replaced by other processes, or a combination thereof.

[0101] Moreover, the scope of the present application is not intended to be limited to the particular embodiments of the process, machine, manufacture, and composition of matter, means, methods and steps described in the specification. As one of ordinary skill in the art will readily appreciate from the present disclosure, processes, machines, manufacture, compositions of matter, means, methods, or steps, presently existing or later to be developed, that perform substantially the same function or achieve substantially the same result as the corresponding embodiments described herein may be utilized according to the present disclosure. Accordingly, the appended claims are intended to include within their scope such processes, machines, manufacture, compositions of matter, means, methods, or steps.

What is claimed is:

1. A manufacturing method, comprising:

disposing at least one semiconductor device and at least one inactive element on a carrier, wherein the at least one inactive element includes a main portion and at least one through via extending through the main portion;

forming an encapsulant on the carrier to encapsulate the at least one semiconductor device and the at least one inactive element;

forming a redistribution structure on a first surface of the encapsulant, a first surface of the at least one inactive element and a first surface of the at least one semiconductor device, wherein the redistribution structure is electrically connected to the at least one semiconductor device and the at least one through via of the at least one inactive element;

disposing an upper electronic device on the redistribution structure; and

removing the carrier.

2. The method of claim 1, further comprising:

forming at least one first bump on the at least one through via; and

forming at least one second bump on a second surface of the at least one semiconductor device.

3. The method of claim 2, further comprising:

dicing the encapsulant and the redistribution structure.

4. The method of claim 3, further comprising:

attaching the at least one first bump and the at least one second bump to a substrate.

5. The method of claim 2, wherein the at least one first bump and the at least one second bump are formed concurrently.

6. The method of claim 1, wherein the upper electronic device is electrically connected to the redistribution structure by hybrid bonding.

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